## Module 2

# Accounting Statements and Finance

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

# Why Do We Need Accounting?

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1. What do firms own?
   - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?
3. How much money did firms make?
   - Income? Cash flow?

# Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?


CNBC, August 18, 2026 (3:50),
"Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

# Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- We will have a clear understanding after this module


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# Three Financial Statements

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  - What do firms own and owe at a point in time

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  - How much did firms earn in the past year

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- **Balance sheet**
  - What do firms own and owe at a point in time
- **Income statement**
  - How much did firms earn in the past year
- **Statement of cash flows**
  - Cash inflows and outflows during the past year

# Securities and Exchange Commission (SEC)

- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure

U.S. Securities and Exchange Commission

SEC.gov | EDGAR

Electronic Data Gathering, Analysis, and Retrieval

sec.gov/edgar/search

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- Distinguish between market and book values
- Explain why income differs from cash flow
- Understand the taxation of corporate and personal income

## Module 2, Part 1

# The Balance Sheet

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$$\textbf{Assets} = \textbf{Liabilities} + \textbf{Equity}$$

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# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current assets**

Assets that can generate return within one year

| Assets | |
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| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
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| Total fixed assets | 29,200 |
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**Current liabilities**

Obligations due within a year

| Liabilities and equity | |
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Most liquid → Least liquid

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| Fixed assets | | Total liabilities | 21,400 |
| Tangible fixed assets | | | |
| Property, plant, equipment | 48,200 | Shareholders' equity | |
| Less accum. depreciation | 19,700 | Paid-in capital | 6,300 |
| Net tangible fixed assets | 28,500 | Retained earnings | 13,600 |
| Intangible asset | 700 | Total shareholders' equity | 19,900 |
| Total fixed assets | 29,200 | | |
| **Total assets** | **41,300** | **Total liabilities and equity** | **41,300** |

Paid first → Paid last

- **Liquidity**, the ease of converting an asset into cash

# Book Value vs. Market Value

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- In general, book $\neq$ market value

# Example, Issuing Stock to Build a Data Center ($ Billions)

| **1. Today** | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

| **2. Issue stock** | | | |
|---|---|---|---|
| Cash | 2 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

| **3. Build data center** | | | |
|---|---|---|---|
| Cash | 0 | Liabilities | 0 |
| Fixed assets | 2 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

- *Should you revise fixed assets if the data center is now worth more than $2 billion?* **No**
- *Should you revise shareholders' equity if someone is willing to buy the business at $4 billion?* **No**

# Book Value vs. Market Value

1. Book value of equity = Book value of assets − Book value of liabilities
   - E.g., Apple at the end of fiscal year 2025,
     - Book value of assets, $359,241 Mn
     - Book value of liabilities, $285,508 Mn
     - What is the book value of equity?

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2. Market value of equity = Market price of each share × Total shares outstanding
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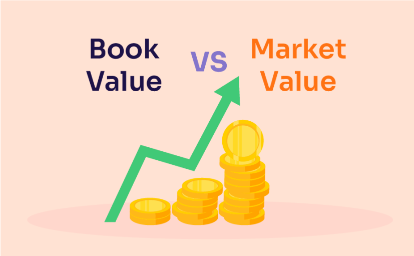
- Book value of equity reflects the money that equity investors have invested into the firm



# Book Value vs. Market Value

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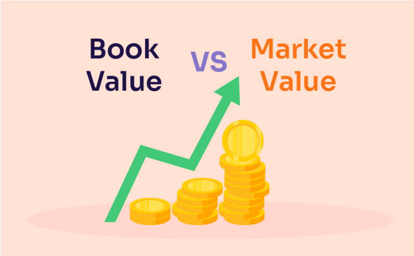
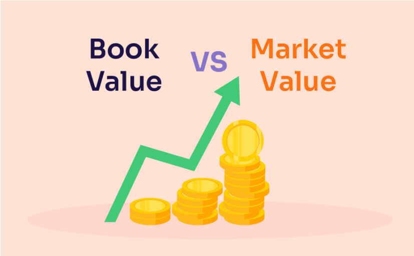
- Market value of equity reflects the money investors expect the firm to make in the future



# Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future
- Naturally, market value of equity $>$ book value of equity for a successful business



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- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?

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- Which of the following are current assets? Select all that apply.

a. Cash.
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c. Wages payable (due in 2 weeks).
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# Recap of Part 1

- Accounting conveys information to investors and stakeholders
- The balance sheet, assets, liabilities, and equity at a point in time
- Assets $=$ Liabilities $+$ Equity
- In general, book $\neq$ market value

## Module 2, Part 2

# The Income Statement

# Income Statement

- **Income statement**, how much firms earned in the past year

# Income Statement

- **Revenues**, sales made in the period, whether or not cash was collected
  - E.g., a firm sells 100 products to customers, each at a price of \$100
  - Revenues $= 100 \times \$100 =$ \$10,000

| | \$ Million |
|---|---:|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Income Statement

- **Operating expenses**, the costs of running the business

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# Income Statement

- **Operating expenses**, the costs of running the business
  - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
    - E.g., each of the 100 products sold costs the firm \$60 to make
    - COGS $= 100 \times \$60 =$ \$6,000

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- This spreading is **depreciation**
- Two things happen with depreciation,
  - In the income statement, subtract depreciation from revenues as a non-cash expense
  - In the **balance sheet**, add to accumulated depreciation

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| Year | 0 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|
| **Cash flow (\$ billions)** | **−2.0** | 0 | 0 | 0 | 0 | 0 | |
| **Effect on net income** | 0 | **−0.4** | **−0.4** | **−0.4** | **−0.4** | **−0.4** | ← Depreciation |

# Income Statement

- **EBIT**, earnings before interest and taxes
  - EBIT = Revenues − COGS − SG&A expenses − Depreciation

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|---|---|
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# Income Statement

- **Interest expense**, the interest paid on the firm's debt for the period

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# Income Statement

- **Taxes**
  - Taxable income = EBIT − Interest expense
  - Taxes = Taxable income × Tax rate

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# Income Statement

- **Net income**, what the business earned in the period after every expense

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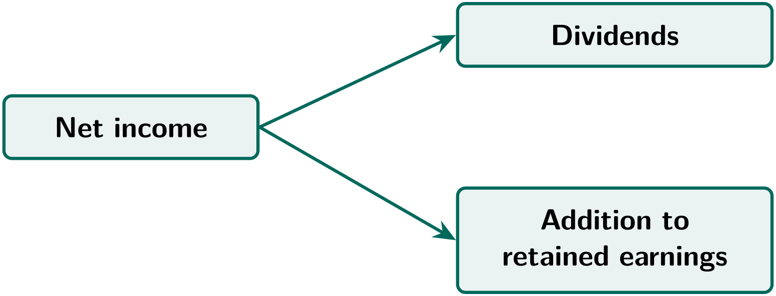

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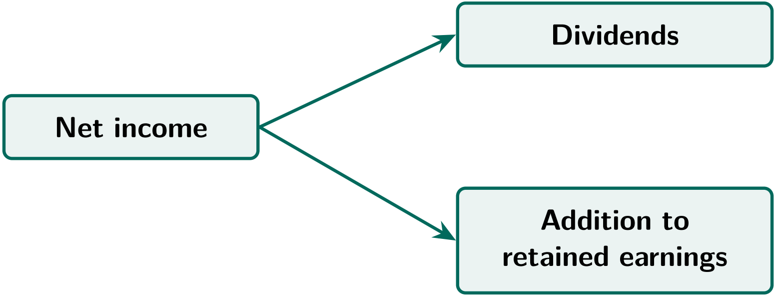


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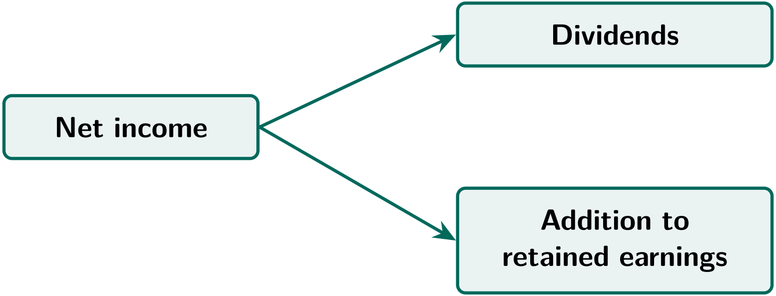


- **Dividends**, cash paid out to shareholders now
- **Retained earnings**, accumulated profits kept in the business and reinvested on the shareholders' behalf

# Income Statement, A Memory Aid

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?


CNBC, August 18, 2026 (3:50),
"Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

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- Profitable on an **EBITDA** basis
  - EBITDA, earnings before interest, taxes, depreciation, and amortization
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- Not profitable on a **net income** basis


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# Practice Problem

| | $ |
|---|---:|
| Revenues | ? |
| COGS | 20,000 |
| SG&A expenses | 10,000 |
| Depreciation | 8,000 |
| **EBIT** | **12,000** |
| Interest expense | ? |
| **Taxable income** | **?** |
| Taxes (tax rate = 21%) | 2,100 |
| **Net income** | **7,900** |
| Dividends | 2,500 |
| Addition to retained earnings | ? |

# Practice Problem

| | $ |
|---|---:|
| Revenues | 50,000 |
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| **Net income** | **7,900** |
| Dividends | 2,500 |
| Addition to retained earnings | 5,400 |

# Recap of Part 2

- The income statement, how much firms earned in the past year
- **Depreciation**, this period's share of a fixed asset's cost
- Net income can be kept in the business, or paid out to shareholders
- Taxes are computed **after** depreciation and interest, never before

## Module 2, Part 3

# The Statement of Cash Flows

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- The statement has three parts
  - **Operating activities**, cash from running the business day to day

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# Statement of Cash Flows

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- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
  - **Operating activities**, cash from running the business day to day
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

# Statement of Cash Flows

| Direct method | Indirect method |
|---|---|
| Lists every cash receipt and payment | Starts from net income and adjusts it |
| More time-consuming to prepare | Faster to prepare |
| Rarely used in practice | **The common choice** |

**Direct** →

**Indirect** →

# Statement of Cash Flows

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- Both methods give the same totals

**Direct** →

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# Statement of Cash Flows

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| More time-consuming to prepare | Faster to prepare |
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

**Direct** →

**Indirect** →

# How Do Net Income and Cash Flow Differ?

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| Year | 0 | 1 | 2 | 3 | 4 | 5 | |
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***When spending on fixed assets, CF = NI − spending on fixed assets***

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***Relationship of cash flow (CF) and net income (NI)?***

***When spending on fixed assets, CF = NI − spending on fixed assets***

***When recording depreciation, CF = NI + depreciation***

# How Do Net Income and Cash Flow Differ?

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| | Effect |
|---|---|
| **Cash flow ($ billions)** | **+1.0** |
| **Effect on net income** | **0** |

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| **Cash flow ($ billions)** | **+1.0** |
| **Effect on net income** | **0** |

- Selling stock is not generating income

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***When raising money, CF = NI + amount raised***

# How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet


OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

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| | Effect |
|---|---|
| **Cash flow (\$ millions)** | **−60** |
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| **Change in inventories** | **+60** |


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***When adding to inventories, CF = NI − change in inventories***

# How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for $100 million
- The customer has not paid yet


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***When firm not yet paid by customers, CF = NI − change in accounts receivable***

# How Do Net Income and Cash Flow Differ?

- The firm owed suppliers $30 million for electronic parts
- The firm pays down the $30 million


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# How Do Net Income and Cash Flow Differ?

- The firm owed suppliers $30 million for electronic parts
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| | Effect |
|---|---|
| **Cash flow ($ millions)** | **−30** |
| **Effect on net income** | **0** |
| **Change in accounts payable** | **−30** |


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- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0


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***When firm pays off suppliers, CF = NI + change in accounts payable***

# Operating Activities

**Operating activities**, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

# Why Subtract Change in Net Working Capital (NWC)?

***When adding to inventories, CF = NI − change in inventories***

***When firm not yet paid by customers, CF = NI − change in accounts receivable***

***When firm pays off suppliers, CF = NI + change in accounts payable***

# Why Subtract Change in Net Working Capital (NWC)?

*When adding to inventories, CF = NI − change in inventories*

*When firm not yet paid by customers, CF = NI − change in accounts receivable*

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*Ignoring other adjustments,*

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$$\textit{When adding to inventories, } CF = NI - \textit{change in inventories}$$
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$$\begin{aligned} CF = NI & \\ & - \text{change in inventories} \\ & - \text{change in accounts receivable} \\ & + \text{change in accounts payable} \end{aligned}$$

$$CF = NI - \text{change in (inventories + accounts receivable − accounts payable)}$$

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*Ignoring other adjustments,*

*CF = NI*
*− change in inventories*
*− change in accounts receivable*
*+ change in accounts payable*

*CF = NI − change in (inventories + accounts receivable − accounts payable)*

*CF = NI − change in net working capital (NWC)*

# Operating Activities

| Cash flow from operating activities | |
|---|---|
| Net income | 3,160 |
| Depreciation | 3,000 |
| Change in net working capital | −1,000 |
| Cash flow from operating activities | 7,160 |

$$3{,}160 + 3{,}000 - (-1{,}000) = 7{,}160$$

# Investing Activities

**Investing activities**, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

# Investing Activities

| Cash flow from investing activities | |
|---|---|
| Cash flow from capital expenditure | −3,000 |
| Cash flow from sales of fixed assets | 200 |
| Cash flow from investing activities | −2,800 |

$$-3{,}000 + 200 = -2{,}800$$

# Financing Activities

**Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Financing activities show cash flows from,

1. Selling new shares $\rightarrow$ Cash $\uparrow$
2. Paying dividends $\rightarrow$ Cash $\downarrow$
3. Taking new debt, either a loan or a bond $\rightarrow$ Cash $\uparrow$
4. Repaying principal amount of old debt $\rightarrow$ Cash $\downarrow$

# Practice Problem

- During the year, a business sells new shares for $120,000 and pays dividends of $30,000. It takes a new loan of $80,000 and repays $50,000 of principal on old debt. How much is its cash flow from financing activities?

# Practice Problem

- During the year, a business sells new shares for \$120,000 and pays dividends of \$30,000. It takes a new loan of \$80,000 and repays \$50,000 of principal on old debt. How much is its cash flow from financing activities?
- Cash flow from financing activities $= \$120{,}000 - \$30{,}000 + \$80{,}000 - \$50{,}000 = \$120{,}000$

# Recap of Part 3

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow $=$ Net income $+$ Depreciation $-$ Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

# Practice Problem

- Which of the following will make cash flow higher than net income for a firm?

a. Increase in inventories.
b. Decrease in accounts payable.
c. Decrease in accounts receivable.
d. Expenditures on fixed assets.

# Practice Problem

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# Practice Problem

- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by $-$\$1,000 million. What is its cash flow from operating activities?

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- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by −\$1,000 million. What is its cash flow from operating activities?
- Cash flow from operating activities = Net income + Depreciation − Change in NWC

# Practice Problem

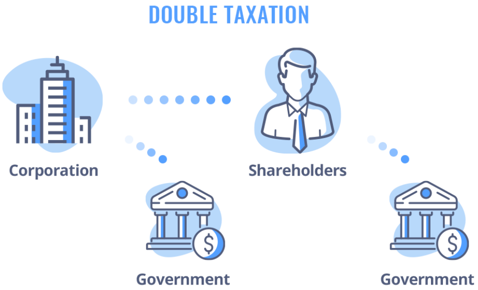
- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by $-$\$1,000 million. What is its cash flow from operating activities?
- Cash flow from operating activities = Net income + Depreciation − Change in NWC
- $\$3{,}160 + \$3{,}000 - (-\$1{,}000) = \$7{,}160$ million

## Module 2, Part 4

# Taxation

# Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income



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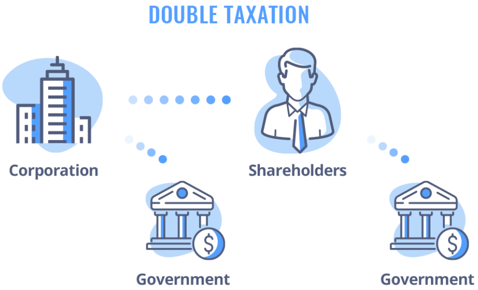
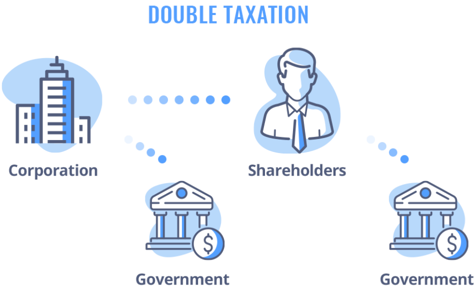


# Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income
  - Shareholders pay tax again on what they receive
- This is **double taxation**



# Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**


WSJ, February 28, 2025 (6:57),
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- **Tax Cuts and Jobs Act (TCJA, 2017)**, cut the rate from 35% to 21%


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# Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic
- **Tax Cuts and Jobs Act (TCJA, 2017)**, cut the rate from 35% to 21%
- The tax cut was extended and largely made permanent in 2025


WSJ, February 28, 2025 (6:57),
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# Interest Tax Shield

- Interest comes first

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
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| Addition to retained earnings | R.E. | **R**e-**E**lections |

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- Interest comes first
- Taxes come last
- Interest reduces corporate taxes

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  - increase the *total* cash flow to investors (debt and equity combined)

# Interest Tax Shield

| Firm A | |
|---|---|
| **EBIT** | **\$5,000** |
| Interest expense | \$1,000 |
| **Taxable income** | **?** |
| Taxes (tax rate = 21%) | ? |
| **Net income** | **?** |

| Firm B | |
|---|---|
| **EBIT** | **\$5,000** |
| Interest expense | \$0 |
| **Taxable income** | **?** |
| Taxes (tax rate = 21%) | ? |
| **Net income** | **?** |

# Interest Tax Shield

| Firm A | |
|---|---|
| **EBIT** | **$5,000** |
| Interest expense | $1,000 |
| **Taxable income** | **$4,000** |
| Taxes (tax rate = 21%) | $840 |
| **Net income** | **$3,160** |

| Firm B | |
|---|---|
| **EBIT** | **$5,000** |
| Interest expense | $0 |
| **Taxable income** | **?** |
| Taxes (tax rate = 21%) | ? |
| **Net income** | **?** |

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| Taxes (tax rate = 21%) | $1,050 |
| **Net income** | **$3,950** |

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**$3,950**

***Investors get $4,160 in total with debt, $3,950 without***

# Interest Tax Shield

- Interest expense can
  - act as a "shield" against corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

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$$\textbf{Interest tax shield} = \textbf{Tax rate} \times \textbf{Interest expense}$$

- In the example, $21\% \times \$1{,}000 = \$210$

# Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by $1

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- So taxes fall by the tax rate $\times$ $1

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$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate})\times\$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\downarrow \text{ by tax rate}\times\$1}$$

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$$\underbrace{\text{Total CF to investors}}_{\uparrow \text{ by tax rate}\times\$1} = \underbrace{\text{Debt investors}}_{\uparrow \text{ by } \$1} + \underbrace{\text{Equity investors}}_{\downarrow \text{ by } (1-\text{tax rate})\times\$1}$$

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- Each dollar of interest raises the *total* cash flow to investors by the tax rate × \$1
- Should a firm take on as much debt as possible? No, bankruptcy costs.

# Practice Problem

- A business pays a 21% tax rate. It takes on debt and pays \$1 more in interest each year. In that year, how does the cash flow to each investor group change?

a. Debt investors get \$1 more, equity investors get \$1 less, no change in total
b. Debt investors get \$1 more, equity investors get \$0.79 less, \$0.21 more in total
c. Debt investors get \$0.79 more, equity investors get \$1 less, \$0.21 less in total
d. Debt investors get \$1 more, equity investors get \$0.21 less, \$0.79 more in total

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- A business pays a 21% tax rate. It takes on debt and pays \$1 more in interest each year. In that year, how does the cash flow to each investor group change?

a. Debt investors get \$1 more, equity investors get \$1 less, no change in total

b. Debt investors get \$1 more, equity investors get \$0.79 less, \$0.21 more in total

c. Debt investors get \$0.79 more, equity investors get \$1 less, \$0.21 less in total

d. Debt investors get \$1 more, equity investors get \$0.21 less, \$0.79 more in total

- Net income falls by $(1 - 21\%) \times \$1 = \$0.79$, and the interest tax shield is $21\% \times \$1 = \$0.21$

# Depreciation Tax Shield

- Depreciation comes first

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Depreciation Tax Shield

- Depreciation comes first
- Taxes come last

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# Depreciation Tax Shield

- Depreciation comes first
- Taxes come last
- Depreciation reduces corporate taxes

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|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
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# Depreciation Tax Shield

- Example, Suppose a business has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | |
|---|---|---|
| Revenues | 80,000 | |
| COGS | 56,000 | |
| SG&A expenses | 16,000 | |
| Depreciation | 3,000 | |
| **EBIT** | **5,000** | |
| Interest expense | 1,000 | |
| **Taxable income** | **4,000** | |
| Taxes | 840 | |
| **Net income** | **3,160** | |

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| **Taxable income** | **4,000** | **3,000** |
| Taxes | 840 | 630 |
| **Net income** | **3,160** | **2,370** |

# Depreciation Tax Shield

- Net income *falls* by $790 million

| $ Million | Baseline | More depreciation | Change |
|---|---|---|---|
| Net income | 3,160 | 2,370 | −790 |
| Add back depreciation | | | |
| **Cash flow** | | | |

# Depreciation Tax Shield

- Net income *falls* by $790 million
- Recall Part 3, Cash flow from operating activities = Net income + Depreciation − Change in NWC

| $ Million | Baseline | More depreciation | Change |
|---|---|---|---|
| Net income | 3,160 | 2,370 | −790 |
| Add back depreciation | 3,000 | 4,000 | +1,000 |
| **Cash flow** | | | |

# Depreciation Tax Shield

- Net income *falls* by \$790 million
- Recall Part 3, Cash flow from operating activities = Net income + Depreciation − Change in NWC
- Cash flow *rises* by \$210 million

| \$ Million | Baseline | More depreciation | Change |
|---|---|---|---|
| Net income | 3,160 | 2,370 | −790 |
| Add back depreciation | 3,000 | 4,000 | +1,000 |
| **Cash flow** | **6,160** | **6,370** | **+210** |

# Depreciation Tax Shield

- Depreciation can also
  - act as a “shield” against corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)
- Depreciation reduces taxable income, cuts taxes, and raises cash flow

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$$\textbf{Depreciation tax shield} = \textbf{Tax rate} \times \textbf{Depreciation}$$

# Math for Depreciation Tax Shield

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- Equity investors get tax rate $\times$ \$1 more
- Each dollar of depreciation raises the *total* cash flow to investors by the tax rate $\times$ \$1
- Can a firm have as much depreciation as possible? No, constrained by cost of acquisition and accounting rules.

# Practice Problem

- A business pays a 21% tax rate. Suppose it had an additional $200 million in depreciation. By how much would its taxes fall? What about its cash flow?

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- Taxes fall by 21% $\times$ \$200 million $=$ \$42 million

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- A business pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its cash flow?
- Taxes fall by $21\% \times \$200$ million $=$ \$42 million
- Cash flow rises by the same \$42 million, the depreciation tax shield

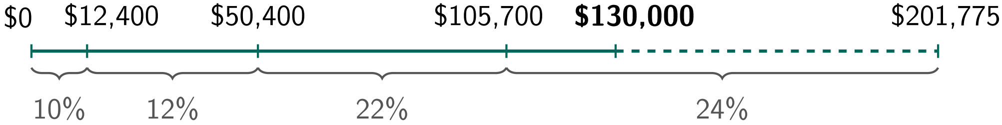
# Personal Taxation

- Personal tax rates matter for salaried employees and non-corporate businesses
- **Marginal tax rate**, the tax rate applied to each extra dollar of income
- **Average tax rate**, total taxes / total income

# Personal Taxation

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| $0–$12,400 | 10 |
| $12,401–$50,400 | 12 |
| $50,401–$105,700 | 22 |
| $105,701–$201,775 | 24 |
| $201,776–$256,225 | 32 |
| $256,226–$640,600 | 35 |
| $640,601 and above | 37 |

Source, IRS Rev. Proc. 2025-32



# Example, Average Tax Rate

Suppose your salary is \$130,000 per year. What would be your average tax rate?

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| \$0–\$12,400 | 10 |
| \$12,401–\$50,400 | 12 |
| \$50,401–\$105,700 | 22 |
| \$105,701–\$201,775 | 24 |
| \$201,776–\$256,225 | 32 |
| \$256,226–\$640,600 | 35 |
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- Total tax $= \$12{,}400 \times 10\% + \$(50{,}400 - 12{,}400) \times 12\% + \$(105{,}700 - 50{,}400) \times 22\% + \$(130{,}000 - 105{,}700) \times 24\% = \$23{,}798$
- Average tax rate $= 23{,}798/130{,}000 = 18.31\%$

# Practice Problem

- Suppose your salary is $30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| $0–$12,400 | 10 |
| $12,401–$50,400 | 12 |
| $50,401–$105,700 | 22 |
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| \$640,601 and above | 37 |

- Total tax $= \$12{,}400 \times 10\% + \$(30{,}000 - 12{,}400) \times 12\% = \$3{,}352$

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- Total tax $= \$12{,}400 \times 10\% + \$(30{,}000 - 12{,}400) \times 12\% = \$3{,}352$
- Average tax rate $= 3{,}352/30{,}000 = 11.17\%$

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- Total tax $= \$12{,}400 \times 10\% + \$(30{,}000 - 12{,}400) \times 12\% = \$3{,}352$
- Average tax rate $= 3{,}352/30{,}000 = 11.17\%$
- Marginal tax rate $= 12\%$

# Recap of Part 4

- The U.S. federal corporate tax rate is **21%**
- Interest and depreciation are deducted (subtracted) before taxes, so both shield income from tax
- A tax shield is worth the tax rate $\times$ the deduction, raising the *total* cash flow to investors
- Marginal rate applies to each extra dollar of income, average rate is total taxes over total income

# Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved

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- Net income $\neq$ cash flow, add back depreciation, subtract the change in NWC
- Interest and depreciation shield taxes, raising the *total* cash flow to investors

# Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?

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- Taxable income = Taxes / Tax rate = \$2,100 / 21% = \$10,000

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- Taxable income = Taxes / Tax rate = \$2,100 / 21% = \$10,000
- Net income = Taxable income − Taxes = \$10,000 − \$2,100 = \$7,900

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- Taxable income = Taxes / Tax rate = \$2,100 / 21% = \$10,000
- Net income = Taxable income − Taxes = \$10,000 − \$2,100 = \$7,900
- EBIT = Taxable income + Interest expense = \$10,000 + \$2,000 = \$12,000

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- Taxable income = Taxes / Tax rate = \$2,100 / 21% = \$10,000
- Net income = Taxable income − Taxes = \$10,000 − \$2,100 = \$7,900
- EBIT = Taxable income + Interest expense = \$10,000 + \$2,000 = \$12,000
- Revenues = EBIT + COGS + SG&A expenses + Depreciation = \$12,000 + \$20,000 + \$10,000 + \$8,000 = \$50,000

# Practice Problem

If a firm's net income is positive and its depreciation is positive, which of the following could account for a negative cash flow from operating activities?

a. Inventories decrease while accounts receivable and accounts payable stay the same.
b. Accounts receivable increases while inventories and accounts payable stay the same.
c. The firm raises money by issuing stocks.
d. A large addition is made to plant and equipment.

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# Thank you!

Please let me know if you have any questions.